

MATA37H3 — Week 11

Answer Key & Misconception Notes

For tutor use only — do not show the student until after their attempt. (Power Series I: Radius/Interval of Convergence & Function Representations)

EXAMPLE 1 — Answer

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{n^2}{(n+1)^2} |x+2| \rightarrow |x+2|; \text{ need } |x+2| < 1, \text{ so } R = 1, \text{ open interval } (-3, -1).$$

At $x = -1$: $\sum \frac{(-1)^n}{n^2}$ converges absolutely. At $x = -3$: $\sum \frac{(-1)^n(-1)^n}{n^2} = \sum \frac{1}{n^2}$ converges.

Interval: $[-3, -1]$.

Common misconception: Forgetting that **both** endpoints can converge — students often assume alternating series behavior at one endpoint means the other must diverge; here $p = 2$ makes both converge absolutely regardless of sign.

EXAMPLE 2 — Answer

Ratio $\rightarrow \frac{|x|}{n+1} \rightarrow 0 < 1$ for every x , so $R = \infty$, interval $(-\infty, \infty)$. No endpoints to check because there are none when $R = \infty$.

Common misconception: Some students still “check $x = \pm\infty$ ” — endpoints only exist (and only need checking) when R is a finite positive number.

EXAMPLE 3 — Answer

$$\left| \frac{a_{n+1}}{a_n} \right| = \frac{(n+1)!}{(n+1)^{n+1}} \cdot \frac{n^n}{n!} |x| = \frac{n^n}{(n+1)^n} |x| = \left(\frac{n}{n+1} \right)^n |x| \rightarrow \frac{1}{e} |x| \text{ (standard limit). Need } \frac{1}{e} |x| < 1 \Leftrightarrow |x| < e. R = e.$$

Common misconception: Treating $\left(\frac{n}{n+1} \right)^n$ as $\rightarrow 1$ instead of $\rightarrow 1/e$ — this is the same limit that defines e in disguise $\left(\left(1 + \frac{1}{n} \right)^{-n} \rightarrow e^{-1} \right)$ and is easy to rush past.

EXAMPLE 4 — Answer

$$\frac{1}{1+3x} = \frac{1}{1-(-3x)} = \sum_{n=0}^{\infty} (-3x)^n = \sum_{n=0}^{\infty} (-3)^n x^n, \text{ valid for } |3x| < 1, \text{ i.e. } \left(-\frac{1}{3}, \frac{1}{3} \right).$$

Common misconception: Substituting $-3x$ but forgetting to raise the -3 to the n th power along with x — writing $(-3)x^n$ instead of $(-3)^n x^n$.

EXAMPLE 5 — Answer

$$\frac{x}{4+x^2} = x \cdot \frac{1}{4} \cdot \frac{1}{1 - \left(-\frac{x^2}{4}\right)} = \frac{x}{4} \sum_{n=0}^{\infty} \left(-\frac{x^2}{4}\right)^n = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{4^{n+1}}, \text{ valid for } \left|\frac{x^2}{4}\right| < 1 \Leftrightarrow |x| < 2. \text{ Interval: } (-2, 2).$$

Common misconception: Forgetting to multiply the whole series by the leftover factor of x at the end, or dropping the $\frac{1}{4}$ pulled out front.

EXAMPLE 6 — Answer

$$\frac{d}{dx} \left[\frac{1}{1-x} \right] = \frac{1}{(1-x)^2}, \text{ and } \frac{d}{dx} \left[\sum_{n=0}^{\infty} x^n \right] = \sum_{n=1}^{\infty} n x^{n-1}, \text{ so } \frac{1}{(1-x)^2} = \sum_{n=1}^{\infty} n x^{n-1} = \sum_{n=0}^{\infty} (n+1) x^n, \text{ valid for } |x| < 1 \text{ (same } R \text{ by Theorem 6.4).}$$

Common misconception: Differentiating the *closed form* correctly but forgetting to reindex the series (leaving it as $\sum n x^{n-1}$ starting from $n = 1$ is fine, but writing $\sum n x^{n-1}$ starting from $n = 0$ silently includes a nonsense term).